Instrument list and I/O count

Flash drum D-400 unit · instrument index, I/O count, transmitter calibration, control valves, SIS separation, loop diagrams

Document no. FD-V1-IN-INL-001

Every instrument of FD-400-PID-001/002 with its service, type, range, normal value, alarm and trip settings, I/O type, system, fail position and location; the I/O count with spares; the calibration of FT-400 and LT-400; the control-valve C_v over the three HMB cases; the separation of the SIS from the BPCS and two loop diagrams. The trips themselves are specified in the C&E FD-V1-IN-CE-001 and the SRS FD-V1-IN-SRS-001.

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Holds and engineering queries

- EI-01 PT-420 range below the PSV set
- EI-02 PSHH-426 and PT-420 tap the same line
- EI-03 FCV-400 rated Cv for choked flow at the maximum rate
- EI-04 SIS reset and bypass switches not on the P&ID
- EI-05 TCV-401 valve ΔP versus the model
- EI-06 Level transmitter calibrated for one density

Note. Numbers come from the simulator model (design case and steady model runs) and the stated assumptions; generated for teaching, not for construction. Engineering queries raised by this document are listed in it and in FD-V1-PM-DOC-001.

Rev	Date	Description	Originator	Checked	Approved
0	28-OCT-26	Issued for teaching	A. Soottitantawat	—	—

DOCUMENT NO.	REV	STATUS	ORIGINATOR	DATE
FD-V1-IN-INL-001	0	Issued for teaching	Apinan Soottitantawat	28-OCT-26

1 · Basis and legend

Source	P&ID FD-400-PID-001 (bubbles, lines) and -002 (control configuration, loop data, SIS C&E); the simulator alarm table ALM; trip constants K.Lshh 90 %, K.Pshh 250 kPa g, K.Lsll 5 %; normal values = HMB FD-V1-PE-HMB-001 design case
I/O types	AI analog input (4–20 mA), AO analog output (4–20 mA to a positioner), DI discrete input (switch, push-button, limit switch), DO discrete output (solenoid); "—" = no wired signal (software function, local or mechanical device)
Systems	DCS basic process control system; SIS safety instrumented system (separate logic solver, SRS); Local (no wire); Mechanical (relief valve)
Fail positions	FC fail closed, FO fail open, "to bypass" = 3-way valve to the bypass port; SIS inputs open their contact on trip and the solenoids de-energise to trip
Recommended additions	rows marked rec. come from the SRS (SIF-1 finding, SIF-3B/4/5 proposals) and the HAZOP (R-01, R-08, R-19, R-20, R-28); they are not counted in the I/O unless approved
Assumed	20 % installed spare; DCS 16 and SIS 8 channels per card; globe valves F_L 0.90, x_T 0.72; γ 1.09 for the drum vapour; remote-seal fill fluid 935 kg/m³

2 · Instrument list

Tag	Service	Type / principle	Loop	Range	Units	Normal	Alarm L	Alarm H	Trip / setting	I/O	System	Fail position	Sheet	Line / location	Note
FT-400	Feed flow	Orifice FE-400 + DP transmitter (√ in the DCS)	FIC-400	0–25	m³/h	16.0	—	—	—	AI	DCS	—	1	2"-P-4001-A1A	range = FIC-400 scale; orifice and DP range in part 4
FIC-400	Feed flow controller	DCS function block	FIC-400	0–25	m³/h	16.0	5.0	20.0	—	—	DCS	—	1, 2	DCS	
FCV-400	Feed let-down valve	Globe, equal %, positioner (flashing-service trim, EP-02)	FIC-400	0–100	%	73.5	—	—	—	AO	DCS	FC	1	2"-P-4002-A1A	C_v in part 5
TT-400	Feed temperature	RTD Pt100 in thermowell	TI-400	0–100	°C	10.0	—	—	—	AI	DCS	—	1	2"-P-4001-A1A	
TI-400	Feed temperature indicator	DCS function block	TI-400	0–100	°C	10.0	—	—	—	—	DCS	—	1, 2	DCS	
TT-401	Hot-feed temperature	RTD Pt100 in thermowell	TIC-401	0–150	°C	74.0	—	—	—	AI	DCS	—	1	2"-P-4002-A1A	
TIC-401	Hot-feed temperature controller	DCS function block	TIC-401	0–150	°C	74.0	69.0	79.0	—	—	DCS	—	1, 2	DCS	
TCV-401	Hot-oil 3-way valve (E-400 port / bypass port)	3-way, linear, positioner	TIC-401 / FIC-405	0–100	% to E-400	30.5	—	—	—	AO	DCS	to bypass	1	2"-HO-4011-B1A	0 % = all hot oil on the bypass
FT-405	Hot oil to E-400	Orifice FE-405 + DP transmitter	FIC-405	0–40	t/h	11.0	—	—	—	AI	DCS	—	1	3"-HO-4013-B1A	
FIC-405	Hot-oil flow controller (Rev B slave of TIC-401)	DCS function block	FIC-405	0–40	t/h	11.0	—	—	—	—	DCS	—	1, 2	DCS	
TT-402	Hot-oil supply temperature	RTD Pt100 in thermowell	TI-402	0–250	°C	180.0	—	—	—	AI	DCS	—	1	2"-HO-4011-B1A	
TI-402	Hot-oil supply temperature indicator	DCS function block	TI-402	0–250	°C	180.0	—	—	—	—	DCS	—	1, 2	DCS	
TT-403	Hot-oil return from E-400	RTD Pt100 in thermowell	TI-403	0–250	°C	120.1	—	—	—	AI	DCS	—	1	3"-HO-4015-B1A	
TI-403	Hot-oil return temperature indicator	DCS function block	TI-403	0–250	°C	120.1	—	—	—	—	DCS	—	1, 2	DCS	
PDT-406	Hot-oil header ΔP (supply − return)	DP transmitter	PDI-406	0–300	kPa	150.0	—	—	—	AI	DCS	—	1	2"-HO-4011-B1A / 2"-HO-4012-B1A	implied by PDI-406 on the P&ID
PDI-406	Hot-oil header ΔP indicator	DCS function block	PDI-406	0–300	kPa	150.0	100.0	200.0	—	—	DCS	—	1, 2	DCS	
LT-400	D-400 level	DP transmitter with remote seals on the bridle	LIC-400	0–100	%	50.0	—	—	—	AI	DCS	—	1	D-400 N6A/B	0–100 % = 0.30–3.30 m above BT; calibration in part 4
LIC-400	D-400 level controller	DCS function block	LIC-400	0–100	%	50.0	20.0	80.0	—	—	DCS	—	1, 2	DCS	
LG-400	D-400 level gauge	Magnetic level gauge on the bridle	—	0–100	%	50.0	—	—	—	—	Local	—	1	D-400 N6A/B	needs no power: the reference when LT-400 is doubted
LCV-400	Bottoms level valve	Globe, equal %, positioner (flashing-service trim, EP-04)	LIC-400 / FIC-410	0–100	%	63.4	—	—	—	AO	DCS	FC	1	2"-P-4003-A1A	
FT-410	Bottoms flow	Orifice FE-410 + DP transmitter	FIC-410	0–25	m³/h	14.0	—	—	—	AI	DCS	—	1	2"-P-4003-A1A	liquid at its bubble point: upstream of LCV-400 (EP-03)
FIC-410	Bottoms flow controller (Rev B slave of LIC-400)	DCS function block	FIC-410	0–25	m³/h	14.0	8.0	18.0	—	—	DCS	—	1, 2	DCS	
AT-410	Bottoms butane	Process GC, liquid sample vaporised at the analyser	AI-410	0–50	wt% C4	26.6	—	—	—	AI	DCS	—	1	2"-P-4003-A1A	sample conditioning SC-410 (heated vaporiser)
AI-410	Bottoms butane indicator (spec ≤ 28 wt%)	DCS function block	AI-410	0–50	wt% C4	26.6	—	28.0	—	—	DCS	—	1, 2	DCS	
TT-410	Flash temperature (drum liquid)	RTD Pt100 in thermowell	TI-410	0–100	°C	49.2	—	—	—	AI	DCS	—	1	2"-P-4003-A1A	
TI-410	Flash temperature indicator	DCS function block	TI-410	0–100	°C	49.2	40.0	60.0	—	—	DCS	—	1, 2	DCS	
PT-420	D-400 pressure	Pressure transmitter	PIC-420	0–300	kPa g	102.7	—	—	—	AI	DCS	—	1	4"-PV-4021-A1A	range does not reach the PSV set (EI-01)
PIC-420	D-400 pressure controller	DCS function block	PIC-420	0–300	kPa g	102.7	70.0	130.0	—	—	DCS	—	1, 2	DCS	
PCV-420	Vapour pressure-control valve	Globe, equal %, positioner	PIC-420	0–100	%	38.7	—	—	—	AO	DCS	FO	1	4"-PV-4021-A1A	
FT-420	Vapour flow	Orifice FE-420 + DP transmitter (P, T compensated)	FI-420	0–1,000	Nm³/h	611.0	—	—	—	AI	DCS	—	1	4"-PV-4021-A1A	implied by FI-420 on the P&ID
FI-420	Vapour flow indicator	DCS function block	FI-420	0–1,000	Nm³/h	611.0	300.0	—	—	—	DCS	—	1, 2	DCS	
AT-420	Vapour butane	Process GC, heat-traced vapour sample line	AI-420	50–100	wt% C4	76.8	—	—	—	AI	DCS	—	1	4"-PV-4021-A1A	sample line traced (EP-09, HT-420)
AI-420	Vapour butane indicator (spec ≥ 75 wt%)	DCS function block	AI-420	50–100	wt% C4	76.8	75.0	—	—	—	DCS	—	1, 2	DCS	
AT-400	Nitrogen in the drum vapour	Thermal-conductivity analyser, heat-traced sample line	AI-400	0–100	mol% N_2	0.0	—	—	—	AI	DCS	—	1	4"-PV-4021-A1A	0–100 % so the start-up from a nitrogen-filled drum stays on scale (draft 0–20 %)
AI-400	N_2 in vapour indicator	DCS function block	AI-400	0–100	mol% N_2	0.0	—	5.0	—	—	DCS	—	1, 2	DCS	
LSHH-416	D-400 level high-high	Level switch on its own nozzle (separate from LT-400)	I-1	0–100	%	50.0	—	—	> 90 % for 5 s	DI	SIS	opens on trip (de-energised)	1, 2	D-400 N7 (3.00 m above BT)	contact closed in normal operation, opens to trip
PSHH-426	D-400 pressure high-high	Pressure switch on its own tap (separate from PT-420)	I-1	0–600	kPa g	102.7	—	—	> 250 kPa g for 3 s	DI	SIS	opens on trip (de-energised)	1, 2	4"-PV-4021-A1A	adjustable range 0–600 kPa g; tap on the same line as PT-420 (EI-02)
LSLL-417	D-400 level low-low	Level switch on its own nozzle	I-2	0–100	%	50.0	—	—	< 5 % for 5 s (XV-418 open)	DI	SIS	opens on trip (de-energised)	1, 2	D-400 N8 (0.45 m above BT)	
HS-400	Emergency shutdown push-button	Hard-wired push-button on the console (latching, guarded)	I-1 (ESD)	—	—	—	—	—	pushed: XV-408, XV-409, XV-418 close	DI	SIS	opens on push	2	Control room	
I-1	SIS function: high level / high pressure / ESD → feed and hot oil off	SIS logic (latched, manual reset)	I-1	—	—	—	—	—	—	—	SIS	—	2	SIS logic solver	C&E FD-V1-IN-CE-001, SRS SIF-1 / SIF-2
I-2	SIS function: low-low level → bottoms off	SIS logic (latched until level > 10 %)	I-2	—	—	—	—	—	—	—	SIS	—	2	SIS logic solver	SRS SIF-3
XV-408	Feed SIS valve	Ball valve, spring-return actuator, tight shut-off	I-1	0–100	open / closed	open	—	—	closes on I-1	—	SIS	FC	1, 2	2"-P-4001-A1A	
XY-408	Solenoid of XV-408	3-way solenoid valve, 24 V DC, energised to open	I-1	—	—	—	—	—	—	DO	SIS	de-energise to trip (vents the actuator)	2	XV-408	
ZSO-408	XV-408 open limit switch	Proximity switch	I-1	—	—	—	—	—	—	DI	SIS	—	2	XV-408	stroke-time check, valve position on the DCS via the SIS link
ZSC-408	XV-408 closed limit switch	Proximity switch	I-1	—	—	—	—	—	—	DI	SIS	—	2	XV-408	
XV-409	Hot-oil SIS valve	Ball valve, spring-return actuator, tight shut-off	I-1	0–100	open / closed	open	—	—	closes on I-1	—	SIS	FC	1, 2	2"-HO-4011-B1A	
XY-409	Solenoid of XV-409	3-way solenoid valve, 24 V DC, energised to open	I-1	—	—	—	—	—	—	DO	SIS	de-energise to trip (vents the actuator)	2	XV-409	
ZSO-409	XV-409 open limit switch	Proximity switch	I-1	—	—	—	—	—	—	DI	SIS	—	2	XV-409	stroke-time check, valve position on the DCS via the SIS link
ZSC-409	XV-409 closed limit switch	Proximity switch	I-1	—	—	—	—	—	—	DI	SIS	—	2	XV-409	
XV-418	Bottoms SIS valve	Ball valve, spring-return actuator, tight shut-off	I-2 and ESD	0–100	open / closed	open	—	—	closes on I-2 and ESD	—	SIS	FC	1, 2	2"-P-4003-A1A	
XY-418	Solenoid of XV-418	3-way solenoid valve, 24 V DC, energised to open	I-2 and ESD	—	—	—	—	—	—	DO	SIS	de-energise to trip (vents the actuator)	2	XV-418	

Tag	Service	Type / principle	Loop	Range	Units	Normal	Alarm L	Alarm H	Trip / setting	I/O	System	Fail position	Sheet	Line / location	Note
ZSO-418	XV-418 open limit switch	Proximity switch	I-2 and ESD	—	—	—	—	—	—	DI	SIS	—	2	XV-418	stroke-time check, valve position on the DCS via the SIS link
ZSC-418	XV-418 closed limit switch	Proximity switch	I-2 and ESD	—	—	—	—	—	—	DI	SIS	—	2	XV-418	
PCV-430	N_2 blanket regulator	Self-acting regulator	—	0–100	kPa g	set 30 kPa g	—	—	—	—	Local	—	1	1"-N-4031-A1A	C_v and restriction orifice open (ER-06)
PSV-400	D-400 relief valve	Conventional spring-loaded, 2J3	—	—	—	set 450 kPa g	—	—	—	—	Mechanical	—	1	3"-PV-4023-A1A	orifice from the Relief tab (ER-01)
LT-400B (rec.)	D-400 level, independent high-level alarm	DP transmitter on its own taps	LAH-400B	0–100	%	50.0	—	85.0	—	AI	DCS	—	—	D-400 (new nozzles)	SRS SIF-1: SIL 1 needs this alarm (HAZOP R-19); set 85 % (assumed)
XY-400 (rec.)	LCV-400 trip solenoid (SIF-3B, 1oo2 final elements)	Solenoid on the LCV-400 actuator	I-2	—	—	—	—	—	—	DO	SIS	de-energise to close LCV-400	—	LCV-400	SRS proposal SIF-3B (HAZOP R-20)
TSHH-404 (rec.)	E-400 hot-feed temperature high-high (SIF-4)	Temperature switch on its own element at the E-400 outlet	SIF-4	0–150	°C	—	—	—	—	DI	SIS	—	—	2"-P-4002-A1A	SRS proposal SIF-4 → XV-409 (HAZOP R-08); tag proposed here
FSLL-407 (rec.)	Feed flow low-low on its own element FE-407 (SIF-5)	Flow switch / transmitter on FE-407	SIF-5	0–25	m³/h	—	—	—	—	DI	SIS	—	—	2"-P-4001-A1A	SRS proposal SIF-5 → XV-409 (HAZOP R-01)
HS-401 (rec.)	SIS reset push-button	Hard-wired push-button on the console	I-1 / I-2	—	—	—	—	—	—	DI	SIS	—	—	Control room	SRS: manual reset at the SIS console (EI-04)
HS-416M (rec.)	Maintenance override switch for LSHH-416	Key switch, status to the DCS	I-1	—	—	—	—	—	—	DI	SIS	—	—	SIS cabinet	SRS bypass under permit, bypass shown on the DCS (HAZOP R-28, EI-04)
HS-426M (rec.)	Maintenance override switch for PSHH-426	Key switch, status to the DCS	I-1	—	—	—	—	—	—	DI	SIS	—	—	SIS cabinet	SRS bypass under permit, bypass shown on the DCS (HAZOP R-28, EI-04)
HS-417M (rec.)	Maintenance override switch for LSLL-417	Key switch, status to the DCS	I-2	—	—	—	—	—	—	DI	SIS	—	—	SIS cabinet	SRS bypass under permit, bypass shown on the DCS (HAZOP R-28, EI-04)

3 · I/O count and spares

Without the recommended additions (the basis of the cabinets and the UPS load)

System	I/O type	Used	Spare (20 %, rounded up)	Installed	Cards	Channels	Free %
DCS	AI	15.0	3.0	18.0	2.0	32.0	53
DCS	AO	4.0	1.0	5.0	1.0	16.0	75
DCS	DI	0.0	0.0	0.0	0.0	0.0	0
DCS	DO	0.0	0.0	0.0	0.0	0.0	0
DCS	total	19.0	4.0	23.0	3.0	48.0	60
SIS	AI	0.0	0.0	0.0	0.0	0.0	0
SIS	AO	0.0	0.0	0.0	0.0	0.0	0
SIS	DI	10.0	2.0	12.0	2.0	16.0	38
SIS	DO	3.0	1.0	4.0	1.0	8.0	62
SIS	total	13.0	3.0	16.0	3.0	24.0	46
Unit	total	32.0	7.0	39.0	6.0	72.0	56

With the recommended additions

System	I/O type	Used	Spare (20 %, rounded up)	Installed	Cards	Channels	Free %
DCS	AI	16.0	4.0	20.0	2.0	32.0	50
DCS	AO	4.0	1.0	5.0	1.0	16.0	75
DCS	DI	0.0	0.0	0.0	0.0	0.0	0
DCS	DO	0.0	0.0	0.0	0.0	0.0	0
DCS	total	20.0	5.0	25.0	3.0	48.0	58
SIS	AI	0.0	0.0	0.0	0.0	0.0	0
SIS	AO	0.0	0.0	0.0	0.0	0.0	0
SIS	DI	16.0	4.0	20.0	3.0	24.0	33
SIS	DO	4.0	1.0	5.0	1.0	8.0	50
SIS	total	20.0	5.0	25.0	4.0	32.0	38
Unit	total	40.0	10.0	50.0	7.0	80.0	50

4 · Transmitter calculations

FT-400 orifice and DP transmitter (ISO 5167, C 0.605 typical)

Range	0–25 m³/h = FIC-400 scale; DP at full scale 25 kPa
Fluid	feed at 10 °C, ρ 638.0 kg/m³, μ 0.255 mPa s; 2" Sch 40, D 52.5 mm
Result	β = 0.717, bore d = 37.6 mm, Re 421,382 at full scale; permanent loss ≈ 11.7 kPa at full scale
Signal	linear in ΔP, square root extracted in the DCS: $Q = Q_{max}\cdot\sqrt{((I-4)/16)}$; low-flow cut-off ≈ 1 m³/h

Q m³/h	% of range	ΔP kPa	% of DP span	mA
5.0	20	1.00	4.0	4.64
10.0	40	4.00	16.0	6.56
16.0	64	10.24	41.0	10.55
20.0	80	16.00	64.0	14.24
25.0	100	25.00	100.0	20.00

LT-400 DP level with remote seals

0 % at 0.3 m, 100 % at 3.30 m above the bottom tangent (span 3 m); taps N6A/B at 0.15 and 3.45 m; LP capillary full of fill fluid ρ_f 935 kg/m³ over 3.30 m, transmitter at the lower tap. $\Delta P = \rho_L g(h - 0.15) + \rho_V g(3.45 - h) - \rho_f g\cdot 3.30$, ρ_L 605.4, ρ_V 4.79 kg/m³. **LRV −29.22 kPa, URV −11.55 kPa** (span 17.67 kPa, elevated zero).

Level %	h above BT m	ΔP kPa	mA
0.0	0.30	−29.22	4.00
25.0	1.05	−24.80	8.00
50.0	1.80	−20.38	12.00
75.0	2.55	−15.97	16.00
100.0	3.30	−11.55	20.00

Liquid in the drum	ρ kg/m³	True 50 % reads %	True 90 % reads %
design liquid	605	50.0	90.0
maximum-rate liquid	605	50.0	90.0
turndown liquid	605	50.0	90.0
cold feed-composition liquid at 28 °C (start-up)	619	51.3	92.2
water at 28 °C (commissioning check)	996	85.8	151.8

5 · Control valves over the three HMB cases

Valve	Case	Flow	P1 → P2 kPa g	ΔP kPa	Pv kPa g / x	Regime	Cv model law	Cv real (choked / IEC)	Rated Cv (model 100 %)	Opening model %	Opening real %	Fail	Char.
FCV-400	Design	17.85 m³/h	550 → 102.7	447	359	choked, flashing	7.38	10.82	18.2	73.5	84.7	FC	eq %
FCV-400	Maximum rate	22.32 m³/h	494 → 102.7	391	359	choked, flashing	9.86	15.30	18.2	82.0	94.9	FC	eq %
FCV-400	Turndown	11.16 m³/h	611 → 102.7	508	359	choked, flashing	4.33	6.08	18.2	57.8	67.8	FC	eq %
LCV-400	Design	14.03 m³/h	133.6 → 50.0	83.5	102.7	choked, flashing	13.81	19.22	48.0	63.4	73.1	FC	eq %
LCV-400	Maximum rate	17.53 m³/h	133.6 → 55.7	77.9	102.7	choked, flashing	17.87	24.01	48.0	71.0	79.6	FC	eq %
LCV-400	Turndown	8.79 m³/h	133.6 → 43.9	89.6	102.7	choked, flashing	8.35	12.03	48.0	48.6	59.3	FC	eq %
PCV-420	Design	1,714 kg/h	102.7 → 30	72.7	x 0.356	sub-critical, Y 0.788	33.6	42.7	271	38.7	45.7	FO	eq %
PCV-420	Maximum rate	2,147 kg/h	102.7 → 30	72.7	x 0.356	sub-critical, Y 0.788	42.1	53.5	271	45.3	52.3	FO	eq %
PCV-420	Turndown	1,063 kg/h	102.7 → 30	72.7	x 0.356	sub-critical, Y 0.788	20.8	26.5	271	24.6	31.7	FO	eq %
TCV-401	Design	10.98 t/h	branch ΔP 150	150	—	linear branch law	11.6	valve alone 31.7	38.0	30.5	30.5	to bypass	linear
TCV-401	Maximum rate	14.69 t/h	branch ΔP 150	150	—	linear branch law	15.5	valve alone 42.5	38.0	40.8	40.8	to bypass	linear
TCV-401	Turndown	6.01 t/h	branch ΔP 150	150	—	linear branch law	6.3	valve alone 17.4	38.0	16.7	16.7	to bypass	linear

Rated C_v = 1316 × the model valve constant (m = C·f(x)·√(ρΔP)). FCV-400 and LCV-400: IEC 60534-2-1 choked $\Delta P_{max} = F_L^2(P_1 - F_F P_v)$, numbers of FD-V1-PE-HMB-001 / the Process engineering tab (EP-02, EP-04). PCV-420: IEC gas sizing with γ 1.09, x_T 0.72, expansion factor Y (the model has Y = 1). TCV-401: the model's linear branch law includes the E-400 shell (EI-05). Proposed rated C_v for an opening ≤ 85 % at the maximum rate on the choked ΔP: FCV-400 32 (EI-03), LCV-400 40 (the model's 48 already meets it).

6 · SIS and BPCS separation

Function	BPCS sensor	SIS sensor	Independent	Final elements (SIS — BPCS)	Note
High level (SIF-1, I-1)	LT-400 on the bridle N6A/B	LSHH-416 on nozzle N7	yes	XV-408, XV-409 (SIS) — FCV-400, TCV-401 (BPCS)	separate nozzles; LG-400 shares the bridle with LT-400 (BPCS side)
High pressure (SIF-2, I-1)	PT-420 on 4"-PV-4021-A1A	PSHH-426 on 4"-PV-4021-A1A	**no**	XV-408, XV-409 (SIS) — PCV-420 (BPCS)	separate taps, same process line: a plugged or closed connection at the vapour nozzle N2 is common to both (EI-02)
Low-low level (SIF-3, I-2)	LT-400 on the bridle N6A/B	LSLL-417 on nozzle N8	yes	XV-418 (SIS) — LCV-400 (BPCS)	SIF-3B would add an LCV-400 trip solenoid (XY-400): the BPCS valve then shares the SIF
ESD (HS-400)	—	HS-400 hard-wired to the SIS	yes	XV-408, XV-409, XV-418	not through the DCS

• Logic solver: SIS logic solver independent of the DCS; scan 0.25 s; I-1 and I-2 latch.

• Final elements: XV-408 (feed), XV-409 (hot oil to E-400), XV-418 (bottoms): spring-return, fail-closed, solenoids de-energise to trip.

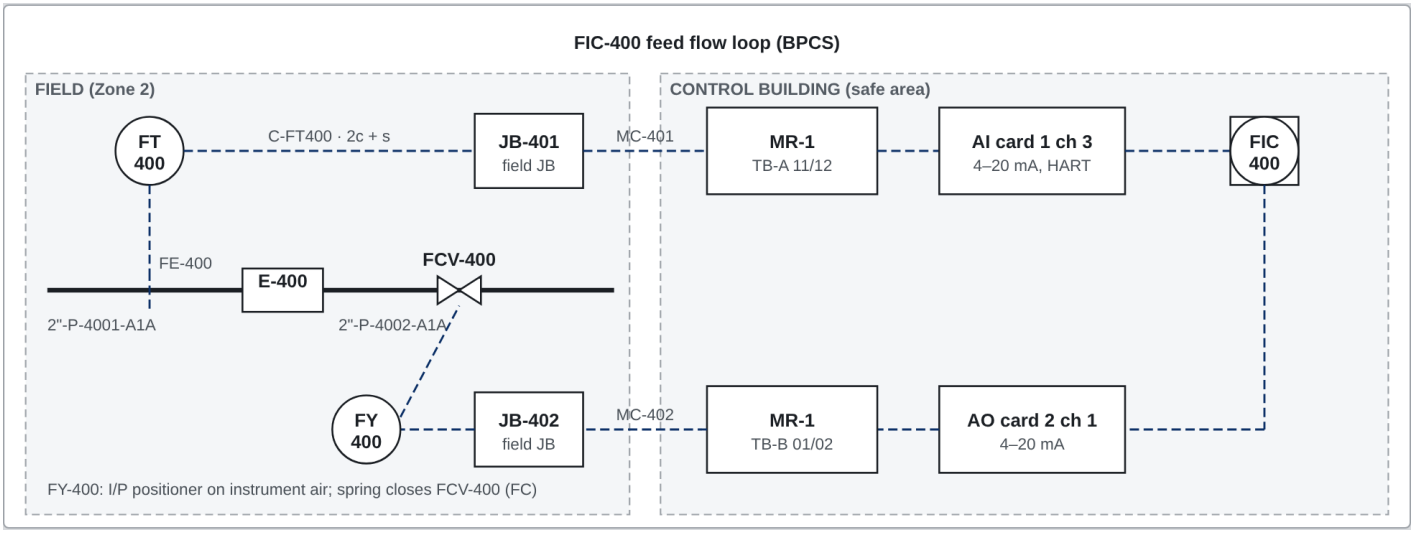
• Energise / de-energise: De-energise to trip; loss of power or instrument air gives the safe state (all XVs closed).

• Bypass and override: Maintenance bypass only under permit with a time limit and compensating measures (operator watching LG-400 and PT-420); bypass status shown on the DCS (HAZOP R-28).

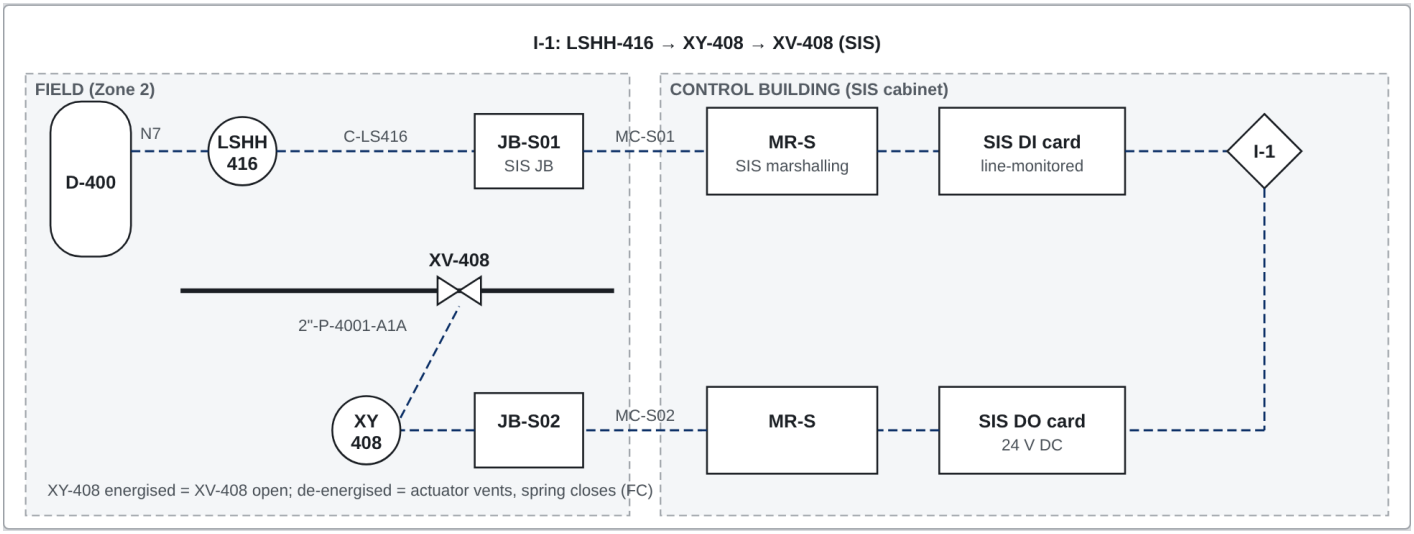
• Reset: Manual reset at the SIS console only when no trip cause is present: level < 85 %, pressure < 230 kPa g; XV-418 opens only with the level above 10 %.

7 · Loop diagrams

FIC-400 feed flow loop (BPCS)



I-1 high level: LSHH-416 → XY-408 → XV-408 (SIS)



Cable, junction-box and terminal numbers are assumed (typical numbering). Field devices Ex ia or Ex d for Zone 2, IIA T3 (FD-V1-EE-LL-001).

8 · Changes to the first draft (epc.json)

Tag	Change	Reason
FT-420, PDT-406, TT-400/402/403/410, AT-400/410/420	Separate transmitter rows added behind the DCS indicators	One row per tag: the transmitter is the I/O, the indicator is software (the draft counted the indicator as the AI)
AT-400	Range 0–20 → 0–100 mol% N_2	The cold drum is full of nitrogen (model cold state 100 %); the start-up step "AI-400 below 2 %" must be readable from 100 % down
XV-408 / 409 / 418	"DO + 2 DI" split into XY (solenoid, DO), ZSO and ZSC (limit switches, DI)	Each wired signal is a tag of its own on the loop drawings and in the I/O count
I-1, I-2	Added as SIS function rows	The trip functions are named in the C&E and the SRS; they hold no I/O
LSHH-416, PSHH-426, LSLL-417	Fail position "contact opens on trip"	De-energise-to-trip (SRS): a broken wire trips, it does not hide a trip
LT-400B, XY-400, TSHH-404, FSLL-407, HS-401, HS-416M/426M/417M	Recommended additions listed (not counted unless chosen)	From the SRS (SIF-1 finding, SIF-3B/4/5 proposals) and the HAZOP (R-19, R-20, R-08, R-01, R-28)

9 · Engineering queries

No.	Subject	Query	Why it matters	Proposed answer	Status
EI-01	PT-420 range below the PSV set	PT-420 / PIC-420 span 0–300 kPa g (controller scale). PSV-400 is set at 450 kPa g and relieves at 495 kPa g (overpressure 10 %); between 300 and the set the DCS shows only "over-range". PSHH-426 is a switch: it gives no reading either.	In an overpressure event the operator loses the pressure trend exactly when it matters; the relief study and the incident review need the peak.	Keep PT-420 0–300 kPa g for control resolution and make PSHH-426 a transmitter 0–600 kPa g (trip in the SIS logic, value shown on the DCS), or range PT-420 0–600 kPa g.	OPEN
EI-02	PSHH-426 and PT-420 tap the same line	PT-420 and PSHH-426 both connect to 4"-PV-4021-A1A downstream of the vapour nozzle N2. Separate taps, but a plugged nozzle, a closed root valve shared by mistake or liquid in the line affects both. The level SIFs use their own drum nozzles (N7, N8) apart from the LT-400 bridle (N6A/B).	IEC 61511 expects the SIF sensor to be independent of the BPCS initiating cause; pressure rising behind a plugged connection would blind the control and the trip together.	Give PSHH-426 its own nozzle on the D-400 top head (add to the nozzle list with V1, EM-05) and its own root valve, car-sealed open.	OPEN
EI-03	FCV-400 rated Cv for choked flow at the maximum rate	Sized on the choked ΔP (EP-02) FCV-400 needs C_v 15.3 at the maximum rate; with the model's rated C_v 18.2 (equal %, R = 30) it would run 94.9 % open, above the 90 % limit. The model, which uses the square-root law over the whole ΔP, shows 82 %.	A valve at its stop at the maximum rate has no control margin left; flashing trims also lose capacity as they wear.	Specify FCV-400 with a flashing-service trim of rated $C_v \geq 32$ (opening ≤ 85 % at the maximum rate, 68 % at design); keep the 2" body and expand after the valve (EP-01).	OPEN
EI-04	SIS reset and bypass switches not on the P&ID	The SRS asks for a manual reset at the SIS console and for maintenance bypasses under permit with the bypass state shown on the DCS (HAZOP R-28). FD-400-PID-002 shows only HS-400.	Without hard-wired reset and override inputs the reset and bypass are done in software, outside the SIS safety manual; the I/O count misses 4 DI.	Add HS-401 (reset) and HS-416M / HS-426M / HS-417M (key-operated overrides with DCS status) to the P&ID and the INL (rows marked rec.).	OPEN
EI-05	TCV-401 valve ΔP versus the model	The model passes hot oil to E-400 in proportion to the TCV-401 opening (linear, 10 kg/s at 100 % with the full 150 kPa header ΔP): C_v 38 for the whole branch. The HMB gives the valve only 20 kPa (H1 → H2, assumed pressures, EP-08), which needs C_v 31.7 for the valve alone at design — the two are not the same valve.	The real 3-way valve and the E-400 shell share the header ΔP; the authority of the valve sets how linear the loop is.	Size TCV-401 with the vendor E-400 shell ΔP and the hot-oil header data; keep the model branch law for teaching and state the difference.	OPEN
EI-06	Level transmitter calibrated for one density	LT-400 (remote seals, 3.3 m between taps) is calibrated with the design liquid 605 kg/m³: LRV -29.22, URV -11.55 kPa. A true 50 % reads 51.3 % with the cold start-up liquid (619 kg/m³) and a true 90 % reads 92.2 %.	At start-up the DCS level is high compared with the gauge; operators must know which one to trust (LG-400 is a float gauge: it also depends on density).	State the calibration density on the datasheet; add a density-compensation note to the start-up procedure (compare LT-400 with LG-400 when the drum is cold).	OPEN